Problem 09: Flood-Safe Corridors

Time limit: 1 seconds

Every monsoon season, several regions across Pakistan face severe flooding. The National Disaster Management Authority (NDMA) wants to plan robust evacuation routes along an existing road network connecting major cities and towns. Some roads are built to modern standards and can safely be used, while others are older and may become unusable during floods. New roads can also be constructed if necessary, but at a cost.

You are given an undirected graph with **n** cities (numbered **1 to n**) and **m** road segments. NDMA has identified a subset of "evacuation cities" – locations such as Karachi, Lahore, Peshawar, Quetta, and others – which must remain mutually reachable by reliable paths even if any single road fails due to flooding.

Formally, a subset **S** of cities is given. NDMA wants to choose a set of roads (some existing, some newly constructed) so that:

1. All cities in **S** lie in a single connected component.

2. Between every pair of distinct cities in **S**, there exist at least two edge-disjoint paths.

3. The total construction/upgrade cost of roads used in the final network is minimized.

Input:

The first line contains three integers **n, m,** and **s ($2 \le n \le 1100$, $1 \le m \le 2000$, $1 \le s \le n$)**: the number of cities, the number of potential road segments, and the number of evacuation cities.

The next line contains s distinct integers **e1, e2, ..., es ($1 \le ei \le n$)**, the labels of the evacuation cities that must be robustly connected.

Each of the next m lines describes a road with four integers **u, v, t,** and **c**:

u and **v ($1 \le u, v \le n$, $u \ne v$)** are the endpoints of the road.

t ($0 \le t \le 2$) is the type of the road:

t = 0: an existing weak road that must be upgraded at cost **c** to be considered reliable.

t = 1: an existing reliable road that can be used at no additional cost; the given **c** should be ignored.

t = 2: a new road that does not currently exist but can be constructed with cost **c**.

c ($1 \le c \le 10^9$) is the upgrade or construction cost for **t = 0** or **t = 2**. For **t = 1**, **c** is present but irrelevant.

Multiple roads between the same pair of cities may exist.
You may choose any subset of roads. The final network consists of all evacuation cities and any other cities you choose to connect with roads, as long as the three conditions above are satisfied using only reliable roads (those chosen, and, if needed, upgraded or built).

Output:

Print a single line.
If it is impossible to satisfy the given constraints print **-1**
Otherwise, print a single integer: the minimum total cost needed to upgrade/build roads so that the evacuation cities satisfy all requirements.

Sample Input 1	Sample Output 1
6 7 3 1 3 5 1 2 1 10 2 3 0 5 3 4 2 4 4 5 1 7 5 6 0 3 2 6 2 6 1 5 2 8	9

Sample Input 2	Sample Output 2
4 3 3 1 2 3 1 2 1 5 2 3 2 7 3 4 2 10	-1